

Lecture 6

Karnaugh Maps

Adjacency and Gray code · two-, three- and four-variable maps ·
prime implicants · product of sums · don't-cares

		1	1
1	1		

$$F = x'y + xy'$$

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 3, §3.1–3.5

Today's roadmap

1

§3.1–3.2

Why a map

Adjacent cells differ in one variable

2

§3.3

Bigger maps

Group 1s in powers of two

3

§3.3

Prime implicants

Essential terms first, then cover the rest

4

§3.5

POS & don't-cares

Group 0s; use X only when it helps

By the end of class you can...

- Label a map in Gray-code order and place minterms in it
- Form the largest legal groups, including wrap-around groups
- Read a product term from any group
- Identify essential prime implicants
- Find a minimum SOP and a minimum POS
- Use don't-care conditions to simplify further

Agenda for today's session

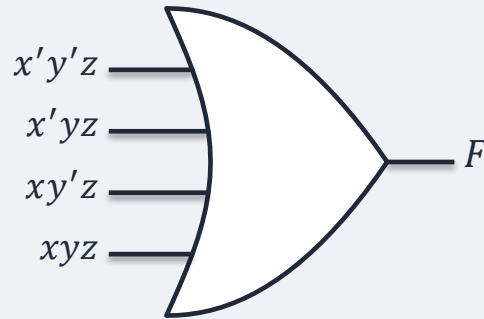
Intro	Why	3-var	Break	4-var	PI/POS	Wrap	
0:00	0:05	0:15	0:40	0:50	1:15	1:40	1:50

Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap, why minimize	—
0:05 – 0:15	Algebra has no stopping rule; the map idea	Discussion
0:15 – 0:40	Two- and three-variable maps, grouping rules	Examples 1, 2 · Your turn 1
0:40 – 0:50	Break	
0:50 – 1:15	Four-variable maps	Example 3 · Your turn 2
1:15 – 1:40	Prime implicants, product of sums, don't-cares	Examples 4–6 · Your turn 3
1:40 – 1:50	Concept check and wrap-up	Homework preview

Why minimize with a map?

§3.1–3.2

The same function, two circuits



Four 3-input ANDs + one 4-input OR (+ inverters)



...is just the wire z : no gates at all.

Algebra works, but...

$$\begin{aligned} F &= x'y'z + x'yz + xy'z + xyz \\ &= x'z(y' + y) + xz(y' + y) = x'z + xz = z \end{aligned}$$

you choose which terms to pair, and nothing tells you when you have reached the minimum.

The map method

Draw the truth table as a grid in which neighbouring cells differ in exactly one variable.

Then every pair of neighbouring 1s is one use of $xy + xy' = x$ — and you can SEE the largest groups.

The map: adjacency by Gray code

§3.1–3.2

Two variables

x \ y	0	1
0	m0	m1
1	m2	m3

Three variables

x \ yz	00	01	11	10
0	m0	m1	m3	m2
1	m4	m5	m7	m6

Gray-code order

Columns run 00, 01, 11, 10 — not 00, 01, 10, 11.

Each step changes one bit, including the wrap from 10 back to 00.

Adjacent means one variable changes

Cells that share an edge are adjacent — and so are the left and right edges of the map, which wrap around.

Note the order of the minterm numbers: m3 sits before m2 because its column is 11.

Grouping rules and reading a group

§3.1–3.2

Legal groups

		yz			
		x	00	01	11
x	0	m0 1	m1 1	m3 1	m2 1
	1	m4 1	m5	m7	m6 1

Copper: the whole top row, $x = 0 \rightarrow x'$

Teal: the four outer cells (edges wrap), $z = 0 \rightarrow z'$

$$F = x' + z'$$

The rules

- 1 Groups hold 1, 2, 4, 8 or 16 cells — a power of two.
- 2 Groups are rectangles; they may wrap around the edges.
- 3 Make every group as large as possible.
- 4 Use as few groups as possible; every 1 must be covered.

Covering a 1 twice is allowed.

Reading a group

A variable that is 1 in every cell of the group: write it (e.g. x).

A variable that is 0 in every cell: write its complement (x').

A variable that changes inside the group: drop it.

Example 1 — A three-variable map

§3.1–3.2

EXAMPLE

Simplify $F(x, y, z) = \Sigma m(2, 3, 4, 5)$.

		yz			
x		00	01	11	10
		m0	m1	m3	m2
0				1	1
1		1	1		
		m4	m5	m7	m6

- 1 Place 1s in cells m2, m3, m4 and m5.
- 2 Look for a group of four: none (the pairs are in different columns).
- 3 Pair m2 (010) with m3 (011).
- 4 $x = 0, y = 1$ stay the same; z changes, so drop it $\rightarrow x'y$
- 5 Pair m4 (100) with m5 (101): $x = 1, y = 0 \rightarrow xy'$
- 6 Every 1 is covered. OR the terms:
- 7 $F = x'y + xy'$ (four literals, and z does not appear)

Example 2 — A wrap-around group

§3.1–3.2

EXAMPLE

Simplify $F(x, y, z) = \Sigma m(3, 4, 6, 7)$.

		yz			
x		00	01	11	10
	0	m0	m1	m3	m2
1		m4	m5	m7	m6
		1		1	1

- 1 Place 1s in m3, m4, m6 and m7.
- 2 Look for a group of four: none forms a rectangle.
- 3 Pair m3 (011) with m7 (111): $y = 1, z = 1 \rightarrow yz$
- 4 m4 and m6 are in the outer columns: adjacent, since the map wraps.
- 5 m4 (100), m6 (110): $x = 1, z = 0$; y changes $\rightarrow xz'$
- 6 Algebra behind the wrap: $xy'z' + xyz' = xz'(y' + y) = xz'$

$$F = yz + xz'$$

Your turn 1 — Three-variable maps

§3.1–3.2

YOUR TURN

≈ 8 min

		yz			
		00	01	11	10
x	0	m0	m1	m3	m2
	1	m4	m5	m7	m6

Use this blank map (or draw your own) for each part.

Find the minimum sum of products.

(a) $F(x, y, z) = \Sigma m(0, 2, 4, 5, 6)$

(b) $F(x, y, z) = \Sigma m(0, 1, 3, 7)$

Hint: in (a), look for a group of four first.

Your turn 1 — Solution

§3.1–3.2

SOLUTION

(a)

		yz			
		00	01	11	10
x	0	m0 1	m1	m3	m2 1
	1	m4 1	m5 1	m7	m6 1

1. Quad m0, m2, m4, m6 (outer columns, wraps)
2. Only $z = 0$ is constant $\rightarrow z'$
3. m5 is left: pair m4 (100), m5 (101) $\rightarrow xy'$
4. OR the terms: $F = z' + xy'$

(b)

		yz			
		00	01	11	10
x	0	m0 1	m1 1	m3 1	m2
	1	m4	m5	m7 1	m6

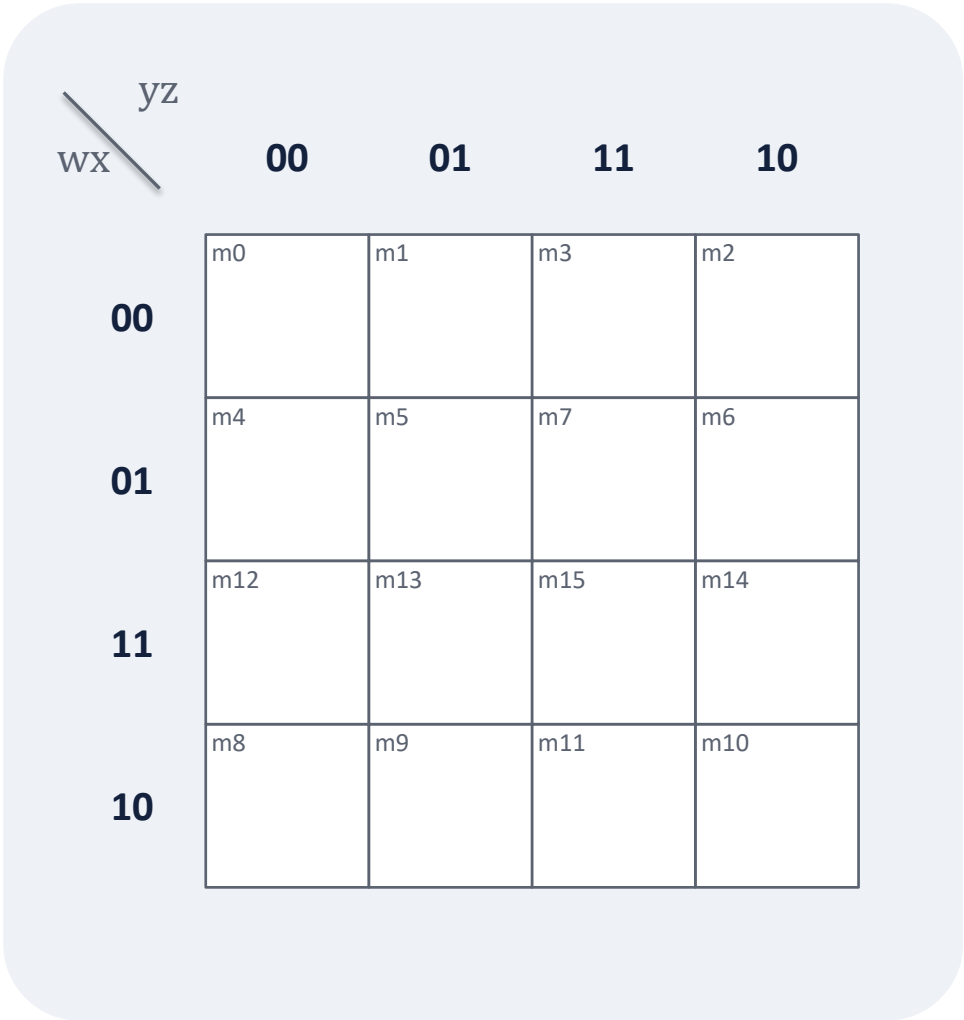
1. No quad. m0 (000), m1 (001) $\rightarrow x'y'$
2. m3 (011), m7 (111) $\rightarrow yz$
3. m1, m3 could pair too, but both are covered.
4. OR the terms: $F = x'y' + yz$

Break

10 minutes — we resume at the 0:50 mark

Next: four-variable maps

The four-variable map



Both directions use Gray code

Rows run 00, 01, 11, 10 — so the bottom row is m8...m10, not m12...m14.

The top and bottom edges wrap, as do left and right: the four corners are one legal group.

Group size	Variables eliminated	Literals left
1 cell	0	4
2 cells	1	3
4 cells	2	2
8 cells	3	1

Example 3 — Eleven minterms

§3.3

EXAMPLE

Simplify $F(w, x, y, z) = \sum m(0,1,2,4,5,6,8,9,12,13,14)$.

$\begin{array}{c} yz \\ \swarrow \searrow \\ wx \end{array}$					
		00	01	11	10
00	m0	1	1		m2 1
01	m4	1	1		m6 1
11	m12	1	1		m14 1
10	m8	1	1		m10

- 1 Place the eleven 1s.
- 2 Largest group first: the octet of the two left columns.
- 3 In the octet only $y = 0$ stays the same $\rightarrow y'$
- 4 Left over: m2, m6 and m14, all in column 10.
- 5 Quad m0, m2, m4, m6 (rows 00–01, outer columns): $w = 0, z = 0 \rightarrow w'z'$
- 6 Quad m4, m6, m12, m14 (rows 01–11, outer columns): $x = 1, z = 0 \rightarrow xz'$
- 7 $F = y' + w'z' + xz'$ (11 minterms $\rightarrow$ 3 terms, 5 literals)

Your turn 2 — Expression to map

§3.3

YOUR TURN

≈ 10 min

CD AB		00	01	11	10
00	m0	m1	m3	m2	
01	m4	m5	m7	m6	
11	m12	m13	m15	m14	
10	m8	m9	m11	m10	

Simplify

$$F = A'B'C' + B'CD' + A'BCD' + AB'C'$$

(a) Expand each term into minterms and place them on the map.

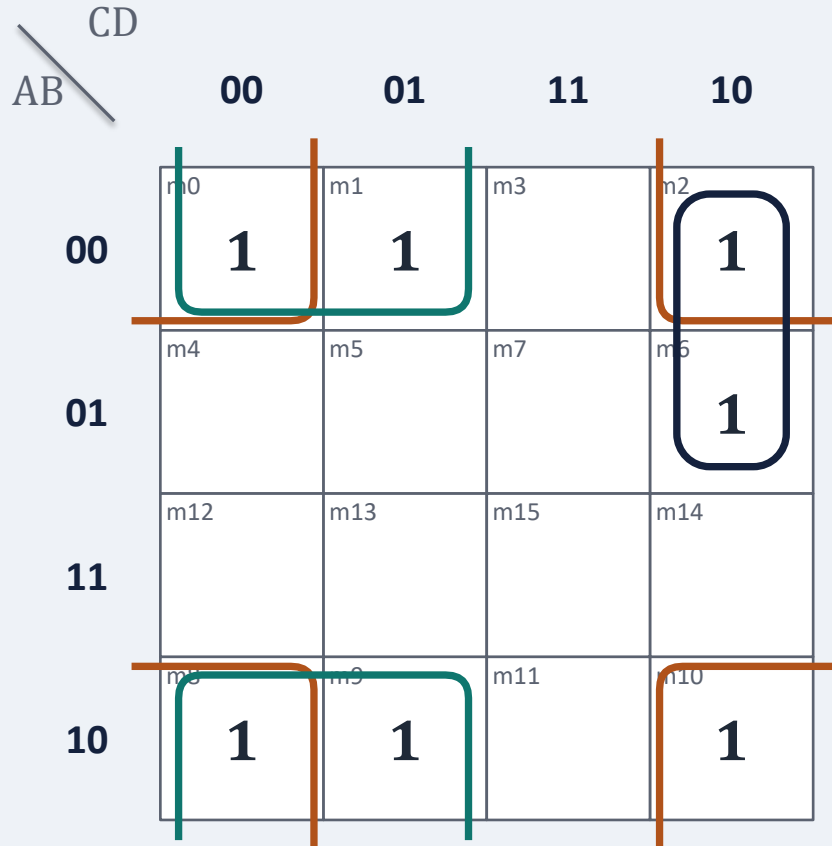
(b) Find the minimum sum of products.

Hint: a term with three literals covers two cells.

Your turn 2 — Solution

§3.3

SOLUTION



- 1 (a) $A'B'C'$ lacks D : $A'B'C'D' + A'B'C'D = m0, m1$
- 2 $B'CD'$ lacks A : $A'B'CD' + AB'CD' = m2, m10$
- 3 $A'BCD'$ is already a minterm: $m6$. $AB'C' \rightarrow m8, m9$
- 4 So $F = \Sigma m(0,1,2,6,8,9,10)$
- 5 (b) Four corners $m0, m2, m8, m10$: $B = 0, D = 0 \rightarrow B'D'$
- 6 $m0, m1, m8, m9$: $B = 0, C = 0 \rightarrow B'C'$
- 7 $m6$ is left: pair with $m2$ ($0010, 0110$) $\rightarrow A'CD'$
- 8 $F = B'D' + B'C' + A'CD'$ (14 literals $\rightarrow 7$)

Example 4 — Essential prime implicants

§3.3

EXAMPLE

Simplify $F(A, B, C, D) = \sum m(0, 2, 3, 5, 7, 8, 9, 10, 11, 13, 15)$.

Essential prime implicants

		CD			
AB \		00	01	11	10
00	m0	1		1	1
01	m4		1	1	
11	m12		1	1	
10	m8	1	1	1	1

Covering the rest

		CD			
AB \		00	01	11	10
00	m0	1		1	1
01	m4		1	1	
11	m12		1	1	
10	m8	1	1	1	1

- 1 **Prime implicant:** a group that cannot be made larger.
- 2 Find 1s that only ONE prime covers.
- 3 m5: only BD covers it $\rightarrow BD$ is **essential**
- 4 m0: only the corners $B'D'$ $\rightarrow$ **essential**
- 5 Still uncovered: m3, m9, m11.
- 6 m3: pick CD (or $B'C$).
- 7 m9, m11: pick AD (or AB').
- 8 $F = BD + B'D' + CD + AD$ — one of four minimum answers

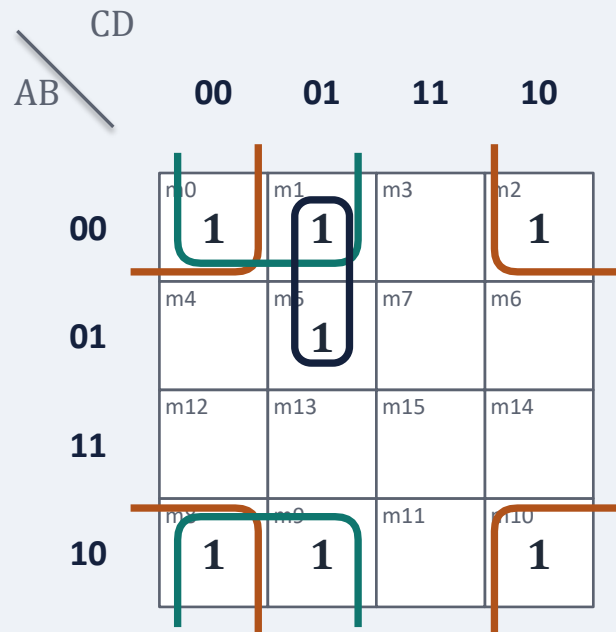
Example 5 — Product of sums

§3.4

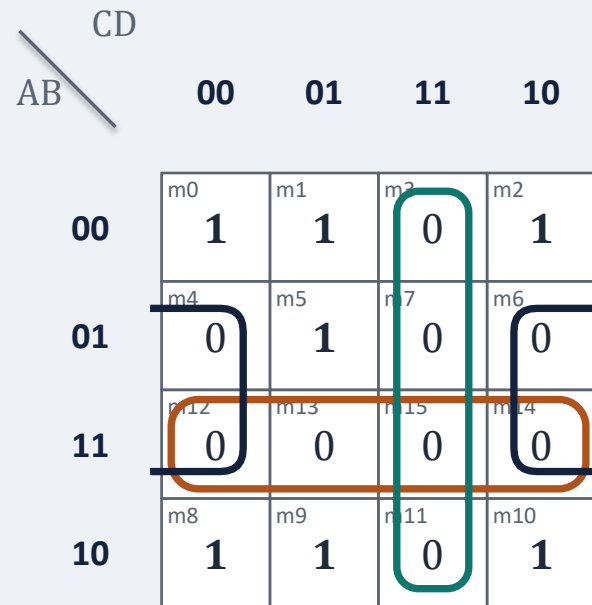
EXAMPLE

Find the minimum SOP and POS of $F(A, B, C, D) = \Sigma m(0, 1, 2, 5, 8, 9, 10)$.

Group the 1s (SOP)



Group the 0s (POS)



1 SOP: group the 1s: $B'D', B'C', A'C'D$

2
$$F = B'D' + B'C' + A'C'D$$

3 POS: write 0s in the empty cells.

4 Group the 0s: AB, CD, BD'

5 That is $F' = AB + CD + BD'$

6 DeMorgan: $F = (AB)'(CD)'(BD')'$

7
$$F = (A' + B')(C' + D')(B' + D)$$

8 Check m5 (0101): both forms give 1 ✓

Example 6 — Don't-care conditions

§3.5

EXAMPLE

Simplify $F(w, x, y, z) = \Sigma m(1, 3, 7, 11, 15)$ with $d(w, x, y, z) = \Sigma m(0, 2, 5)$.

$\begin{array}{c} yz \\ \swarrow \searrow \\ wx \end{array}$					
		00	01	11	10
wx	00	m0 X	m1 1	m3 1	m2 X
	01	m4	m5 X	m7 1	m6
	11	m12	m13	m15 1	m14
	10	m8	m9	m11 1	m10

- 1 A don't-care (X) never occurs: we may treat it as 0 or 1.
- 2 Place 1s in m1, m3, m7, m11, m15 and X in m0, m2, m5.
- 3 Column $yz = 11$: m3, m7, m11, m15: $y = 1, z = 1 \rightarrow yz$
- 4 m1 is left. Treat the X at m0 and m2 as 1.
- 5 The top row is now a quad: $w = 0, x = 0 \rightarrow w'x'$
- 6 m5 is not needed: leave that X as 0.

$$F = yz + w'x'$$

Your turn 3 — A BCD function

§3.5

YOUR TURN

≈ 8 min

CD AB		00	01	11	10
00	m0	m1	m3	m2	
01	m4	m5	m7	m6	
11	m12 X	m13 X	m15 X	m14 X	
10	m8	m9	m11 X	m10 X	

The input $ABCD$ is a BCD digit, so the codes 10–15 never occur.

$$F = \Sigma m(0,2,4,5,6) + d(10,11,12,13,14,15)$$

Find the minimum sum of products.

Watch out: m8 is not a don't-care.

Your turn 3 — Solution

§3.5

SOLUTION

CD						
AB						
		00	01	11	10	
00	m0	1	m1	m3	m2	1
01	m4	1	1	m7	m6	1
11	m12	X	X	m15	m14	X
10	m8		m9	m11	m10	X

1 Place 1s in m0, m2, m4, m5, m6 and X in m10–m15; m8 is 0.

2 Quad m0, m2, m4, m6: $A = 0, D = 0 \rightarrow A'D'$

3 m5 is left. Treat the X at m12 and m13 as 1.

4 Quad m4, m5, m12, m13: $B = 1, C = 0 \rightarrow BC'$

5 Trap: D' alone would also cover m8, which must be 0.

6

$$F = A'D' + BC'$$

Concept check

1

Why do the columns of a map run 00, 01, 11, 10?

2

Can a group contain 6 cells? Can it contain the four corners?

3

What makes a prime implicant essential?

4

How do you get a POS from the map, and when is it cheaper?

Key takeaways

Gray-code order

00 01 11 10

Group sizes

1, 2, 4, 8, 16

A group of 2^k cells

drops k variables

Essential PI

only cover of some 1

POS

group 0s $\rightarrow F' \rightarrow$ DeMorgan

Don't-care X

use only if it helps

The map gives a guaranteed minimum for up to four variables.

Homework #3 due at the start of Lecture 7

This afternoon's lab: Lab B — Verilog and the Nexys 4 DDR: switch to LED

Next: NAND and NOR implementations and the XOR function